

THE LIVING MAP

Spatial Memory for Emergency Robots

TSYP14 Technical Challenge | IEEE RAS x IEEE AESS Tunisia Section Chapters

Phase 1 Proposal A two-robot system that gives a GPS-denied, disconnected environment a persistent memory using autonomous radio beacons.

1. Challenge and motivation

In disaster response, robots may operate inside collapsed, fire-damaged, underground or contaminated environments where GPS and communication are unavailable. The challenge is not only to explore such spaces, but to preserve critical knowledge so that a later robot can continue the mission without starting from zero.

Our proposed Phase 1 working scenario is a Fire / Hazardous Building. The Writer Robot enters first, explores without a pre-existing map, detects safety-relevant conditions, and leaves information behind in beacons. The information can later be recovered through the Outside Network and used to brief an Executor Robot.

2. Proposed system in one view

Subsystem	Hardware	Role / status
Writer Robot	Arduino UNO Q; RPLIDAR; Modulino Movement (LSM6DSOX IMU); Modulino Distance (VL53L4CD ToF); servo + beacon dispenser; thermal sensor candidate MLX90640 or AMG8833; wheeled base, encoders, motor driver, battery and emergency stop to be finalized.	Autonomous exploration, localization, sensing, event detection, beacon decision and log. Core board/sensors confirmed by user; final mechanical and thermal choices open.
Beacon Network	ESP32-class MCU; SX1276/RFM95-class LoRa transceiver; non-volatile memory; LiPo battery + regulator; deployment switch; status LED; antenna. Reference architecture.	Persistent spatial memory and RF broadcast. Beacon-to-beacon relay is proposed; RF band, exact module, protocol and range remain open.
Outside Network Area	Raspberry Pi gateway + GPS receiver; local network interface; MQTT/server layer; PC/browser for command-post map; wireless or satellite backhaul.	RECEIVE, TRANSLATE, CARRY, BRIEF. Architecture designed; final gateway/backhaul hardware open.
Command Post	PC/laptop + display/browser for map/dashboard.	Visualize GPS-referenced events/path and select the mission. Proposed.
Executor Robot	Proposed reuse: UNO Q; RPLIDAR; Modulino Movement/IMU; floor-distance sensing; selected thermal sensor; beacon-compatible RF receiver; drive base + encoders + motor driver; battery; optional transmitter.	Enter later, listen to beacons, verify information, navigate and execute the mission. Architecture designed; hardware not finalized.

Core idea If the Writer fails, the robot may disappear - but the knowledge already stored in deployed beacons remains.

3. Design principles

- GPS-free inside: the Writer and Executor operate in a private coordinate frame referenced to the entrance.
- Persistent memory: beacons stay active after deployment and carry compact WHAT / WHERE / WHEN information.
- No direct inside-to-command-post link: all outside communication passes through the Outside Network Area.
- Resilience by design: Writer log recovery is used when the Writer returns; deployed beacons remain the fallback when it does not.
- Honest prototyping: simulation results are separated from physical hardware validation and open engineering decisions.

4. Writer Robot - entering and exploring

The Writer is the first robot. It starts at the entrance, without a pre-existing map. The current implementation uses a hardware abstraction layer so the decision logic is independent from the physical hardware interface. Planned hardware: Arduino UNO Q, RPLIDAR, Modulino Movement (LSM6DSOX IMU), Modulino Distance (VL53L4CD ToF), servo-based beacon dispenser, and a thermal sensor to be selected (MLX90640 or AMG8833 are current candidates). A wheeled drive base, wheel encoders, motor driver, battery and emergency stop are also planned, with exact parts still to be finalized.

Writer operating stages

Stage	What the Writer does
1. Calibrate	Stops at the entrance for IMU bias calibration, then starts autonomous exploration.
2. Explore	Uses LiDAR-based wall following, obstacle checks and local position estimation.
3. Observe	Reads LiDAR, thermal and downward distance information to detect relevant events.
4. Decide	Scores event candidates and decides whether a beacon is worth consuming.
5. Drop	Builds the beacon message, writes/configures the beacon, releases it and confirms the drop.
6. Continue / Recover	Continues exploration or executes a stuck-recovery maneuver when needed.
7. Return / Upload	Returns using a stored breadcrumb path when the mission ends, then uploads the detailed log near the entrance.

5. Navigation and localization

- Right-hand wall following: maintain approximately 0.40 m from the right wall; slow down while turning and react to front blockage or a missing floor.
- Movement estimate: wheel odometry + IMU heading. The simulation adds gyro bias, encoder scale error and measurement noise.
- Heading correction: a straight wall detected by LiDAR is used as a simplified stand-in for scan matching. The current simulation assumes an axis-aligned, Manhattan-style building.
- Return path: the Writer stores a breadcrumb trail and erases loops so the stored route remains a short known path back to the entrance.

6. Writer state machine

State	Purpose
CALIBRATE	Initial sensor calibration.
EXPLORE	Navigation + sensing + event evaluation.
DROP	Stop, configure, release and confirm beacon.
RECOVER	Back up and turn after a stall.
RETURN	Retrace the stored route to the entrance.
UPLOAD	Send the detailed log to the Outside Network.
DONE	Mission completed.
SAFE_STOP	Safety state for tip-over or unrecoverable conditions.

Implementation boundary The Writer simulation is the most mature software component. The physical RPLIDAR, drive base, thermal sensor, beacon radio and dispenser remain to be integrated and validated.

7. Event detection and beacon decision

The challenge requires at least two event types. Our Phase 1 design uses two primary safety events plus a navigation landmark: HAZARD (heat/fire hotspot), HOLE (collapsed floor), and JUNCTION (branching point).

Type	Detection method	Why it matters	Beacon content in current simulation
HAZARD	Thermal reading reaches the configured heat threshold; the Writer tracks the peak.	Preserves the location of a possible fire/heat source and its severity in the Writer log.	Type, bearing, estimated distance, time.
HOLE	Down-facing distance sensor detects a floor discontinuity above the configured threshold.	Prevents the Executor from blindly crossing a dangerous missing-floor area.	Type, direction, time.
JUNCTION	LiDAR identifies multiple open directions and confirms the condition across consecutive scans.	Preserves a navigation landmark and available exits for the inherited map.	Type, direction, time, 4-bit exits mask.

8. Which beacon is dropped?

The Writer treats beacon stock as a limited resource. Each candidate event receives a score based on event importance, severity, novelty and remaining stock. When stock decreases, the acceptance threshold increases, so high-value events are protected while low-value relay/waypoint candidates can be skipped.

Example candidate	Current policy
HAZARD	High priority; severity increases the score.
HOLE	High fixed priority because it is safety-critical.
JUNCTION	Medium priority; useful as a structural landmark.
WAYPOINT	Lower priority; used when beacon spacing would otherwise break the proposed communication chain.

9. Beacon message and deployment sequence

Field	Meaning
id	Unique beacon identifier.
type	HAZARD / HOLE / JUNCTION / WAYPOINT.
dir_deg	Direction in the private frame.
dist_m	Estimated event distance for HAZARD/HOLE; for JUNCTION/WAYPOINT, the current simulation uses an assumed 3.5 m spacing placeholder.
t	Timestamp used for aging.
ver	Message version for later updates.
exits	4-bit open-direction mask for junctions.

Deployment: detect event -> score -> create message -> configure beacon -> servo release -> confirm physical drop -> beacon becomes active.

Physical beacon reference architecture (proposed): ESP32-class controller, non-volatile memory, independent battery/regulator, deployment switch, status LED, antenna and SX1276/RFM95-class LoRa transceiver. Exact RF technology, frequency, antenna and byte-level protocol remain team decisions.

Design note: neighboring-beacon previous/next relationships and relay toward the Outside Network are proposed resilience upgrades, not implemented in the current Phase 1 simulation. They must be agreed with the team and checked against the challenge interpretation before hardware implementation.

10. Beacon Network - persistent spatial memory

Proposed architecture only. The communication modes below are final-system options; they are not implemented in the current Phase 1 software repository.

Mode	Flow	Purpose
Direct	Beacon -> Outside Network	Fast delivery when a deployed beacon can reach the gateway. Proposed; subject to organizer interpretation of the disconnected-zone premise.
Relay	Beacon -> Beacon -> ... -> Outside Network	Extends communication into disconnected areas. Proposed.
Executor	Beacon -> Executor	Provides local navigation and event information when the second robot enters.

11. Beacon chain proposal

At deployment time, a Writer cannot know the future location of the next beacon. We therefore propose that each newly deployed beacon identifies its predecessor, and that the previous beacon learns the successor through a short-range/local RF exchange. This creates a chain such as B1 -> B2 -> B3 without requiring future positions to be known in advance.

Stored metadata	Purpose
previous_id / next_id	Identifies neighboring beacons in the proposed chain.
direction / distance to neighbor	Supports local navigation between anchors.
message_id / hop_count / TTL	Prevents repeated forwarding and infinite relaying.
version	Allows a later update to supersede older information.

Open design decision: the earlier briefs assumed the Outside Network builds the chain from the Writer log; the local-RF predecessor/successor mechanism is a new proposal. The team must choose or combine these approaches and define the radio write method.

12. Outside Network Area

The Outside Network is the bridge between the disconnected interior and the outside world. Proposed hardware is a Raspberry Pi gateway with a GPS receiver at the entrance, a local network interface and MQTT/server services, plus a PC/browser at the command post for the map. The final wireless or satellite backhaul remains to be selected.

Role	Input	Output
RECEIVE	Writer log I3 (current Phase 1 interface).	Validated stored data. Recovered-beacon information is a future extension.
TRANSLATE	Private coordinates + entrance GPS + building bearing.	GPS-referenced events/path + uncertainty.
CARRY	GPS-referenced data.	Command-post update; store-and-forward if link is down.
BRIEF	Mission + translated data + proposed beacon chain.	Executor briefing I5.

13. Frame translation

The Writer uses a private frame whose origin is the entrance. The Outside Network combines the entrance GPS fix and the building entry bearing to transform each point into East/North displacement and then latitude/longitude. The map should preserve an uncertainty estimate because dead-reckoning and bearing errors grow with distance.

Communication rule: the command post never connects directly to the robots. Inside-to-outside communication passes through the Outside Network Area.

14. Executor Robot - entering with inherited knowledge

The Executor is the second robot. Proposed hardware reuses the Writer architecture where practical: UNO Q, RPLIDAR, Modulino Movement/IMU, floor-distance sensing, the selected thermal sensor, a beacon-compatible RF receiver, drive base with encoders and motor driver, battery, and an optional transmitter. Exact Executor hardware is not yet finalized.

State	Behaviour
BRIEFED	Receives mission, beacon chain, known hazards, clock information and trust rules.
ENTER	Starts from the same entrance convention and establishes its private frame.
LISTEN	Moves toward the expected next beacon while monitoring its own sensors.
TRUST-CHECK	Evaluates beacon age, version and agreement with onboard sensors.
FOLLOW	Uses the beacon chain for local navigation.
ACT	Performs the mission at the target, e.g. reach a heat hazard for an extinguishing task. This is a design target, not a completed Phase 1 execution.
EXIT	Leaves the building by retracing its route; exact exit logic remains to be finalized.
SAFE_STOP	Stops for unrecoverable or unsafe conditions.

15. Trust by age

Beacon status	Executor interpretation
Fresh	Trust more strongly and follow at normal speed.
Aging	Slow down and verify with own sensors.
Stale	Treat as guidance only; rely primarily on onboard sensing.
Contradicted / suspect	Flag and discard the conflicting information.

The challenge presentation gives illustrative fresh/aging/stale examples. Exact thresholds and responsibility for computing age are still team decisions and are not implemented in the current simulation.

16. End-to-end mission scenario - proposed final architecture

Design target, not an implemented end-to-end simulation. Steps 1-4 are represented in the current Writer simulation. Steps involving beacon networking, the Outside Network, command post and Executor are design targets for subsequent implementation.

- 1. Writer enters a fire-damaged building with no GPS and explores autonomously.
- 2. LiDAR detects a junction; the Writer records available exits and may place a JUNCTION beacon.
- 3. The thermal sensor identifies a heat hotspot; the Writer records HAZARD information and places a beacon when the policy accepts it.
- 4. A missing-floor condition is detected; the Writer stops/turns away and records a HOLE event.
- 5. Deployed beacons remain active; direct delivery and multi-hop relay toward the Outside Network are proposed mechanisms.
- 6. If the Writer returns, its detailed log is received by the Outside Network and translated to GPS; if it fails, persistent beacon memory remains the fallback.
- 7. The command post selects the mission and the Outside Network briefs the Executor.
- 8. The Executor enters, listens for beacons, checks age and sensor consistency, follows the inherited route, reaches the target and exits.

Resilience mechanism: two complementary memory paths are targeted - the detailed Writer log when the Writer returns, and persistent beacon memory when the Writer does not.

17. Phase 1 simulation and evidence

The current Writer simulation is implemented in Python and models a hazardous building with LiDAR, thermal sensing, a floor sensor, odometry/gyro imperfections, battery drain, beacon decisions and logging. The simulation is hardware-independent through the WriterHAL interface. The simulated interfaces correspond to the planned LiDAR, thermal, floor-distance, IMU/odometry, battery and beacon-dispenser functions; the physical components are not yet integrated.

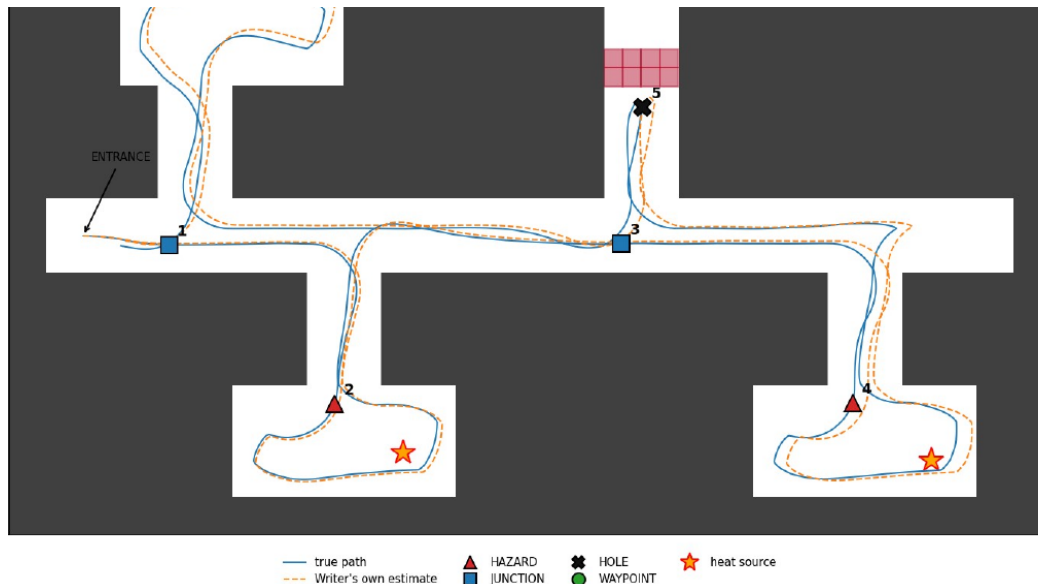


Figure 1. Representative Writer simulation output from the repository; the clean evidence table below uses seed 2.

Current evidence	Observed result	Interpretation
Normal run (seed 2)	Returned; 56.0 m path; 5 beacons; log uploaded; 0.03 m drift; 92% battery.	Clean Writer demonstration.
14-seed execution test	14/14 returned and detected HAZARD + HOLE + JUNCTION by the script criterion; coverage checks still found incomplete/long runs in some seeds.	Passes the stated criterion, not full robustness.
Failure tests	Writer loss, drop failure, hidden debris, low battery and forced upload failure were executed and recorded.	Reproducible Phase 1 evidence.

18. Failure cases and implementation plan

Failure	Observed / expected response	Phase
Writer failure / loss	Writer stops; deployed beacons remain; no log upload.	Phase 1 evidence
Beacon/drop failure	Retry at p=1; no beacon deposited; Writer returns and uploads.	Phase 1 evidence
Hidden debris	RECOVER -> EXPLORE -> return and upload.	Phase 1 evidence
Low battery	RETURN at 30% reserve -> UPLOAD -> DONE; 29% remaining.	Phase 1 evidence
Upload failure	Forced failure; three attempts; upload_failed; log kept on board.	Phase 1 evidence
Outside link loss	Queue/retry is a design scenario; Outside Network code is not implemented.	Design / later
Now - Phase 1		After submission - Final Phase
Freeze I1-I5 interfaces and event set		Finalize RF technology and beacon electronics
Integrate Writer log -> Outside Network -> Executor simulation		Build physical Writer / Beacon / Outside Network / Executor
Generate clean demo, failure evidence and final diagrams		Validate navigation, RF range, GPS translation and trust/aging
Finalize GitHub + <=6 page report		Complete prototype, user manual and final pitch

Known limitations: Some seeds show incomplete room coverage, false stuck detection, long trajectories, and missed/duplicated junction or hazard detections. Heading correction assumes axis-aligned walls; RF propagation, beacon aging and physical integration are not implemented. Status: Writer = IMPLEMENTED in simulation; Beacon, Outside Network, live map and Executor = DESIGN ONLY.